

# No Evidence for Inhibition Control or Interlingual Homograph Interference in Bilingual Sentence Processing: A Self-Paced Reading Study

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## 1. Motivation

- Interlingual homographs (IH)** - Identical orthography but different meanings

- Different meanings → Interference during comprehension

- Inhibition control is one of the mechanisms that suppresses interference [1]



- Pivneva et al. [2] → ↑ Inhibition control ⇒ ↓ Interference, but no good control over experimental materials

- This work reexamines the role of inhibition control by:
  - tighter controls over experimental materials
  - a more reliable inhibition-control battery [3]
  - testing inhibitory aftereffects through self-paced reading of later-presented **translation equivalents (TE)** [4]

## 2. Materials and Design

**German native speakers** with English as L2 (N = 76, based on a priori power analysis) participated in the **Self-Paced Reading (SPR)** experiment

**24 IH-TE** pairs with their controls (matched in length and lexical frequency), **C1** and **C2**

### SPR

| Condition      | P1   | P2     |
|----------------|------|--------|
| <b>IH + TE</b> | GIFT | POISON |
| <b>C1 + TE</b> | CARD | POISON |
| <b>IH + C2</b> | GIFT | JUICE  |
| <b>C1 + C2</b> | CARD | JUICE  |

“PAUL WAS SEARCHING FOR THE \_\_\_\_\_ (P1) WHEN HE NOTICED SOME \_\_\_\_\_ (P2) SPILLED UNDER THE SMALL MICROSCOPE.”

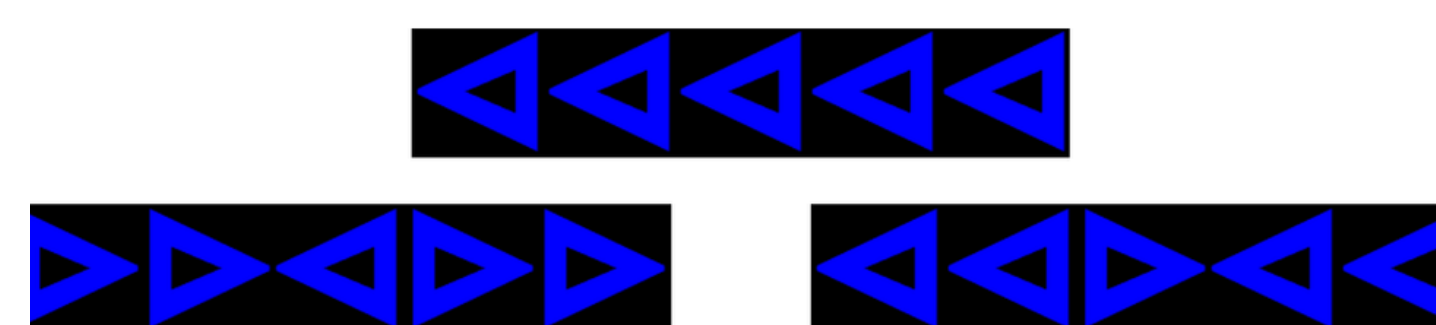
### Inhibition Control Battery

Stroop Squared

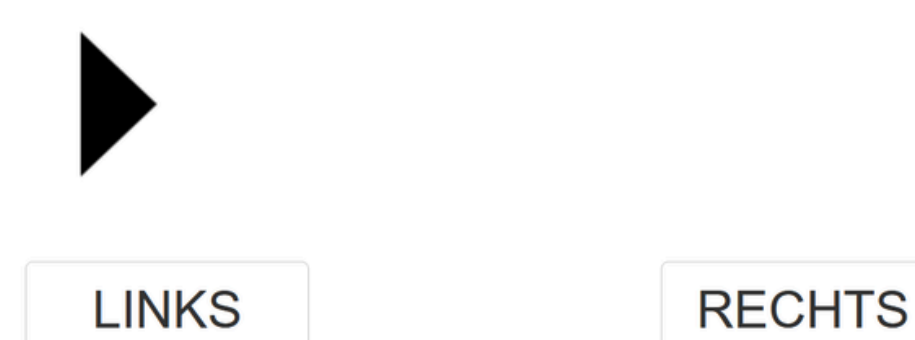
Remaining Time  Point: 2

ROT  
BLAU ROT

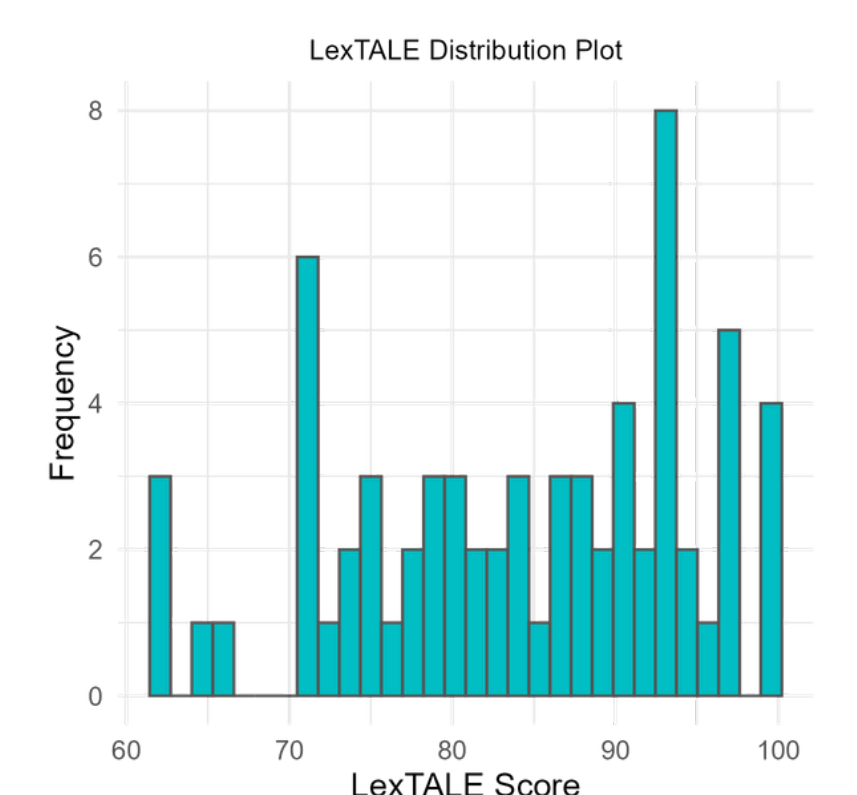
Flanker Squared



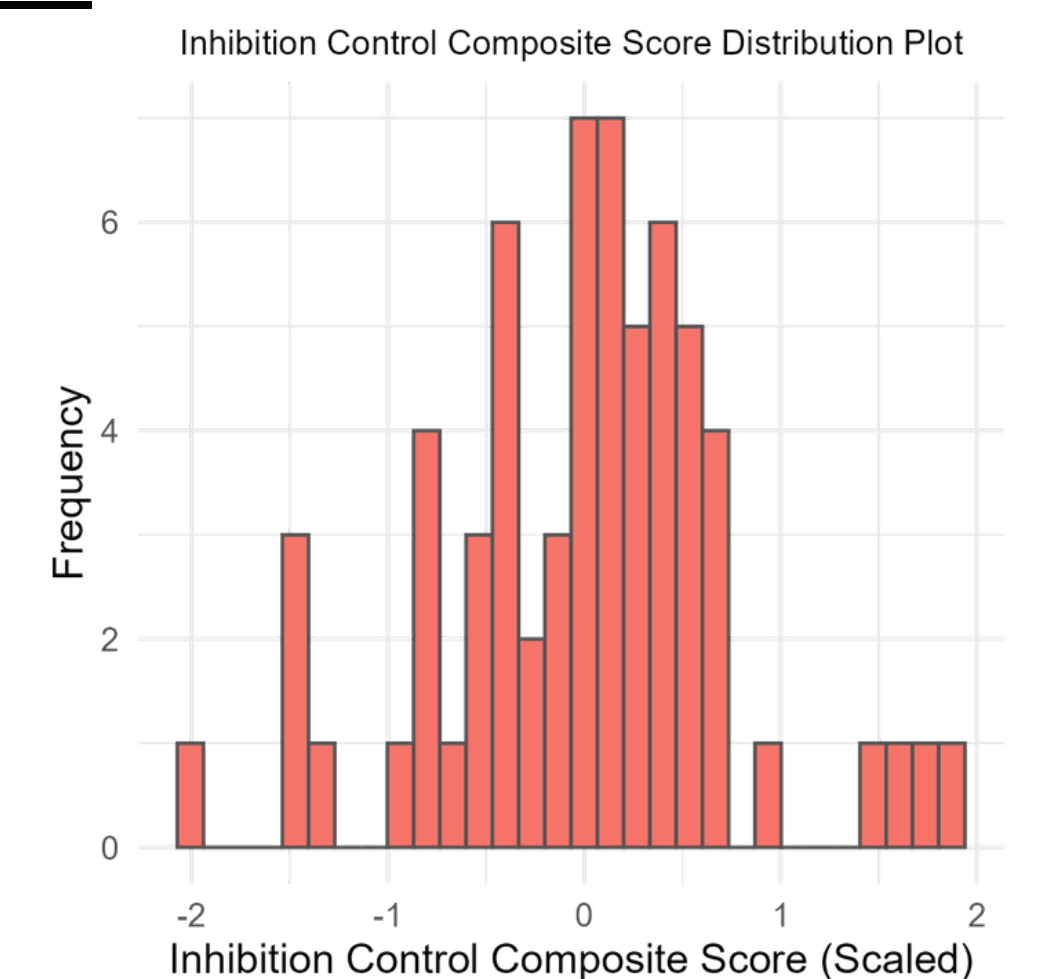
Simon Squared



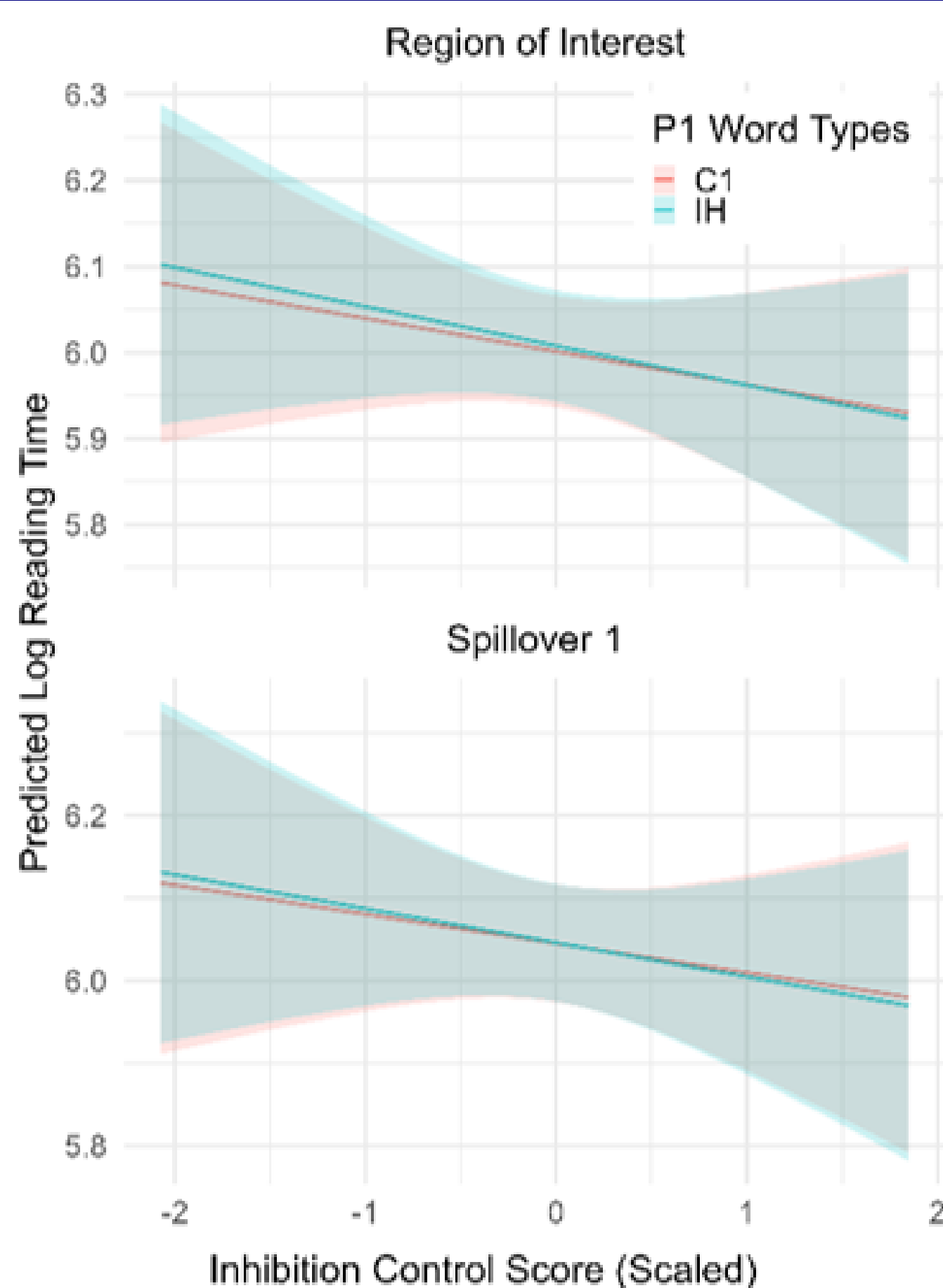
### English Proficiency from LEAP-Q and LexTALE



### Composite Score of Inhibition Control

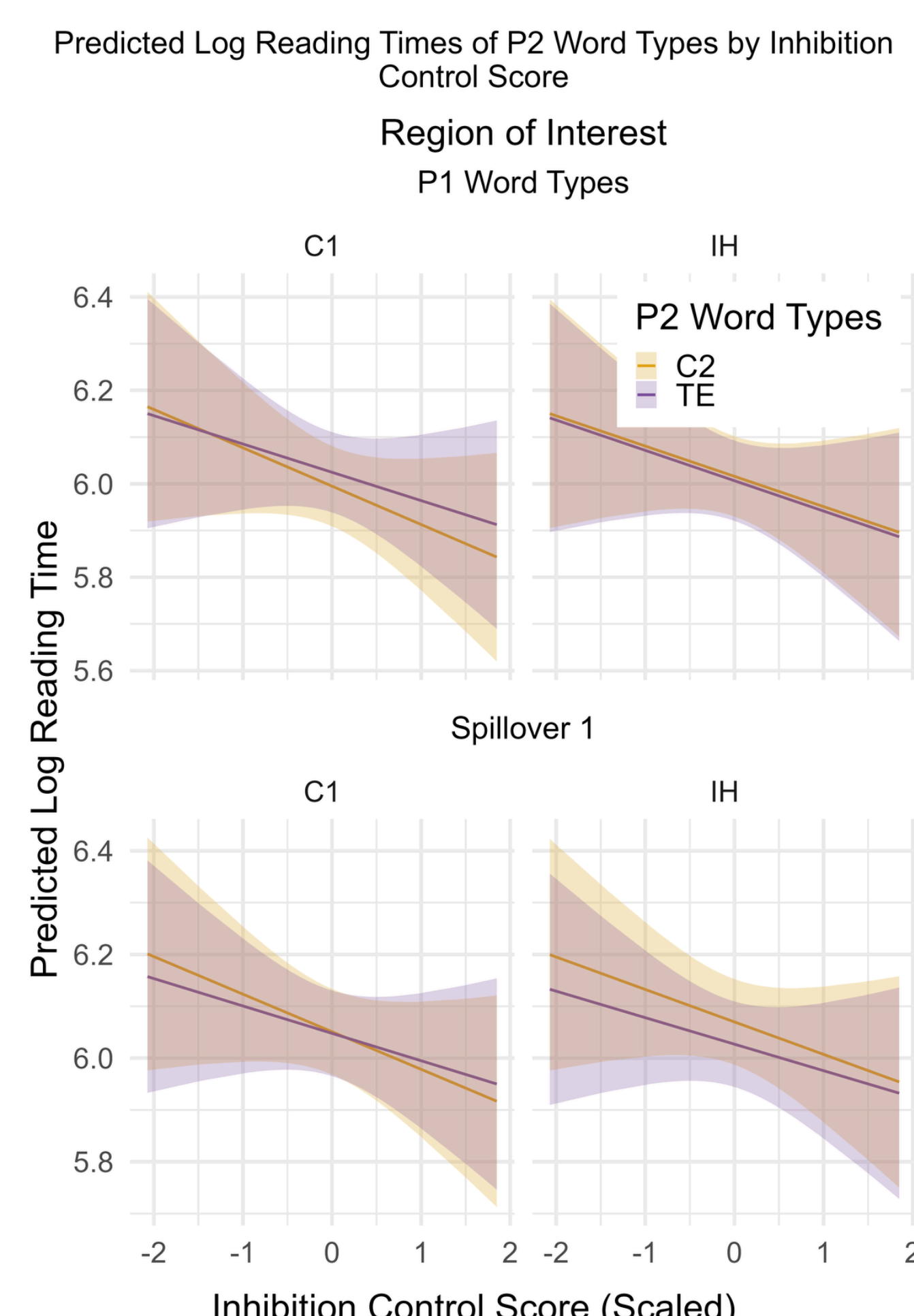


## 3. Do bilinguals experience cross-linguistic interference when processing IH, and is it modulated by inhibition control?



- The selected linear mixed-effects (LME) model had random effects for items and participants
- NO EVIDENCE FOR INTERFERENCE** and **INHIBITION** in reading times of **IH** vs its control, **C1**
- NO EVIDENCE FOR INDIVIDUAL DIFFERENCES** in inhibition control for reading times of **IH** vs **C1**
- Typical effects of word length and word frequency were significant

## 4. Does inhibition during IH processing have inhibitory after-effects for subsequent lexical processing?



- The selected LME model had random effects for items and participants
- NO EVIDENCE FOR INHIBITORY AFTEREFFECTS** in reading times of **TE** when preceded by **IH** vs **C1**
- NO EVIDENCE FOR INDIVIDUAL DIFFERENCES** in inhibitory aftereffects in reading times of **TE** when preceded by **IH** vs **C1**
- Typical effects of word length and word frequency were significant

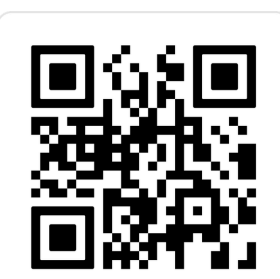
## 5. Why No Evidence for Interference and Inhibition Control?

- Participant variability in interference patterns. Almost half the participants had overall **negative interference costs for IH**
- In pairs like *Gymnasium-Bungalow*, **RT of Bungalow > RT of Gymnasium** → **Corpus frequency** may not be reflective of **vocabulary knowledge for L2 scenarios**
- Absence of inhibitory aftereffects could be due to:
  - Design differences** between this work and [3], and the associated periods of effect decays
  - Participants relied on “**good-enough**” inhibition of alternative meanings instead
- SPR** → Group-level effects ✓ but **Individual differences** ✗ [5]
- Eye-tracking and event-related potentials with carefully controlled materials can shed more light on our observations

## References

- [1] Green, D. W. (1998). Mental control of the bilingual lexico-semantic system. *Bilingualism: Language and Cognition*, 1(2), 67-81.
- [2] Pivneva, I., Mercier, J., & Titone, D. Executive control modulates cross-language lexical activation during L2 reading: evidence from eye movements. *JEP: Learning, Memory, and Cogn.* 40, 787, 2014.
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- [4] Macizo, P., Bajo, T., & Martín, M. C. Inhibitory processes in bilingual language comprehension: Evidence from Spanish-English interlexical homographs. *JML*, 63(2), 232-244, 2010.
- [5] Frinzel, F. F., & Christiansen, M. H. (2024). Capturing individual differences in sentence processing: How reliable is the self-paced reading task?. *Behavior Research Methods*, 56(6), 6248-6257.

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